

# Understanding Compositional Meaning in Biology

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**ABSTRACT.** Recent publications in biology enthusiastically argue that primate vocal call systems possess non-trivial compositionality as their prominent feature, thus paralleling human language. No proposal for an internal computational system explaining this property has been offered so far. However, conceptual and quantitative analysis shows the use of non-human primate compositional meaning to be neither productive nor rule based. Furthermore, formal properties of (higher-order) Markov chains imply only string concatenation, explaining why hierarchical structure, hence structure dependence, is absent from animal call systems. Also, mere acoustic identity of constituent elements with unit calls does not suffice for compositional meaning. In conclusion, higher-order Markov chains and distributional semantics seem to be questionable methods for establishing and explaining compositionality. A humanlike combinatorial capacity cannot be unambiguously demonstrated for non-human primates.

**Key words** primate vocalization, Markov chain, hierarchical structure, compositional meaning, structure dependence, recursion, distributional semantics, monoid vs magma

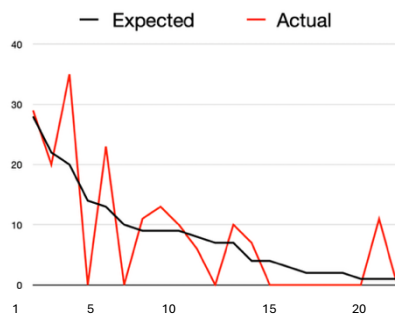
**1. Introduction.** Cases of a single AB combination of two calls A and B are widely attested in animal call systems across various species of birds and primates. This is an interesting finding in itself. Even more interestingly, a recent *Science* study (Berthet et al., 2025) of wild bonobos at the Kokolopori Bonobo Reserve, DRC, found seven bonobo calls with nineteen attested combinations of them (out of a total of 42 logically possible ones). Four are claimed to have compositional meaning, and three of these have non-trivial (“non-intersective”) compositionality. The authors suggest that “compositionality is a prominent feature of the bonobo vocal system, revealing stronger parallels with human language than previously thought.” A smattering of success in identifying and characterizing properties of an internal call system that would allow non-human primates to generate “(non)intersective” combinations would undeniably be a matter of considerable interest.

**2. Conceptual considerations.** Compositional meaning of a complex linguistic expression is (fully) determined by the meanings of its constituents and the way they are combined by syntactic rule. Compositionality is therefore an interpretive property of (hierarchical) structure, not a rule of structure building. Furthermore, since human language is recursive, there’s no non-arbitrary limit to the complexity of its hierarchical structure (Chomsky, 2021). However, while there’s no upper bound on hierarchical complexity, the simplest hierarchical structure requires three primitives (Everaert et al., 2017; Bolhuis et al., 2018). Consequently, we can demonstrate without even directly invoking compositionality that the bonobo call system holds no relevance for the ontogeny/phylogeny of human language. This is because non-trivial compositionality requires hierarchical structure, which is missing from non-human primate communication systems. Compositional meaning is non-trivial (“non-intersective”) if it is not additive/conjunctive. However, non-intersective compositionality of AB is not conjunctive precisely because AB is structurally either  $[[A X Y] B]$  or  $[A [B X Y]]$ . Hierarchical structure is implied. In fact, the authors’ example of non-intersective compositionality itself, “bad dancer,” presupposes hierarchical structure, i.e.,  $[[bad dance] -er]$ , representing “a person who dances badly,” not “a bad person who dances,” which may be represented as  $[bad [dance -er]]$ . <sup>1)</sup>

For hierarchical structure to emerge, at least three primitives are needed that in principle allow for two competing structures, namely  $(a(bc))$  and  $((ab)c)$ , each with a different meaning. These structural ambiguities, which are *productive* in human language, *must* exist by virtue of its hierarchical structure. Hierarchical structure is present at all levels of grammar. In derivational morphology, there's the contrast  $((un\ lock)\ able)$  “can be unlocked” vs.  $(un\ (lock\ able))$  “cannot be locked”; in compounds, we have  $((American\ football)\ player)$  “player of American football” vs.  $(American\ (football\ player))$  “football player who is an American citizen”; and in syntax,  $((kept\ his\ bike)\ in\ the\ attic)$  “kept it in the attic, i.e., his bike” contrasts with  $(kept\ (his\ bike\ in\ the\ attic))$  “kept it, i.e., his bike in the attic.” Compositional contrasts like these are absent in animal call systems for a principled reason, the absence of structure-building rules that apply recursively. What we see here in the bonobo vocal system (and everywhere else in animal call systems), is a *limited* number of *paired* call combinations. Even the simplest hierarchical structure is thus completely absent. There's no structure-building operation that can be recursively applied to yield compositionally relevant hierarchical structures. There may not even be a rule that would generate these combinations but rather memorization of two- and three-call combinations. Furthermore, since the internal program of human language yields unbounded structured expressions (Chomsky, 2021; Berwick & Chomsky, 2016), a finding about some bounded case of output (i.e., all studies of animal call communication thus far) tells us little, if at all, about the core properties of the human internal program.

Little can be inferred simply because the intersection of the computational properties of biologically driven human language (i.e., a magma algebra with a merge operation) and animal call systems (i.e., a monoid algebra with concatenation as its operation) is empty. There's no overlap. Merge is commutative and non-associative while concatenation is associative and non-commutative (Marcolli et al., 2025). Even the calls are different from the word-like primitives of human language (Chomsky, 2013). The language-specific conditions of theta-theory and event semantics (“who does what to whom”) that are part of lexical structure are missing from calls. And if we compare paired bonobo calls with, e.g., noun-noun combinations of human language along the lines of Yang (2013), we expect to see, first, that human language is “consistent with a productive grammar rather than memorization of specific word combinations,” and second, that the primate call combinations “[do] not show the expected productivity of a rule-based grammar” (Yang, 2013). Expectations are borne out.

**3. Empirical motivations.** Assuming a fully statistically independent manner of composition as shown to be the case for human noun-noun compounding, the expectations can be empirically confirmed. First, we compute *expected* probabilities of the binary call combinations  $P(AB)$  from the product of the marginal probabilities of the individual calls  $A$  and  $B$ , i.e.,  $P(AB) = P(A) * P(B)$ . The graphic plots of the *collected frequencies* are then compared against the *expected probabilities* for primate paired calls, as was done for human noun-noun



combinations. Comparison yields different outcomes. Yang's simulation tests using a Monte Carlo method indicate that unlike the human case, where the plots closely match, for the non-human primates, the graphic plot of expected probabilities deviates significantly from the plot of actual frequencies. See the text figure to the left, which plots the distribution of attested structures (red) against a frequency-rank distribution of expected bigrams (black). Abstracting from positional restrictions ( $A\_B, B\_A$ ), the frequency-rank distributions of 21 expected and 11 attested bigrams in the text figure can be more explicitly represented in the figure below. Significantly, the frequency rankings of attested against expected bigrams do not converge.

|                 | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10 | 11 | 12 | 13 | ... | 20 | 21 |
|-----------------|----|----|----|----|----|----|----|----|----|----|----|----|----|-----|----|----|
| <u>expected</u> | 28 | 22 | 20 | 14 | 13 | 10 | 9  | 9  | 9  | 8  | 7  | 7  | 4  | ... | 1  | 1  |
| <u>attested</u> | 29 | 20 | 35 | 0  | 23 | 0  | 11 | 13 | 10 | 6  | 0  | 10 | 7  | 0   | 11 | 0  |
|                 | 2  | 4  | 1  |    | 3  |    | 6  | 5  | 8  | 11 |    | 9  | 10 |     | 7  |    |

In the simulation, the 3<sup>rd</sup> ranked *expected* bigram ‘Peep-yelp\_Yelp’ with frequency 20 corresponded to the 1<sup>st</sup> ranked *attested* bigram with frequency 35; the 20<sup>th</sup> ranked *expected* bigram ‘High-hoot\_Low-hoot’ with frequency 1 corresponded to the 7<sup>st</sup> ranked *attested* bigram with frequency 11. Conversely, the 4<sup>th</sup>, 6<sup>th</sup> and 11<sup>th</sup> ranked *expected* bigrams (‘High-hoot\_Peep’, ‘High-hoot\_Yelp’, ‘Whistle\_Yelp’) with frequencies 14, 10, and 7 respectively, had no *attested* bigrams. [Test data from Charles Yang, pc].

Compositional meaning is clearly not rule-based in the bonobo vocal system. Not only is there no recursive rule, no rule that yields hierarchical structure for compositional meaning, but even bigram composition differs significantly from human compounding. Chomsky’s well-argued position that syntactic competences between humans and non-human primates are sharply discontinuous along different dimensions appears strongly confirmed rather than firmly weakened. So far, there has been no unambiguous demonstration of a humanlike combinatorial capacity in non-human primates.

We plainly need to know more about the structure of these calls (e.g., are the two positions open slots to be filled freely, or slots constrained by categorial or other restrictions?). Even so, we might pursue a slightly different calculation based on the *Tolerance Principle* of Yang (2016) saying that a rule R applicable to N items is productive if and only if the number of exceptions it tolerates is at most  $N/\ln N$ . The study found 7 single calls and 19 distinct binary combinations of these. From their Fig. 3 and Table S1, we infer that only 2 (out of 7) A-calls, i.e., *Peep*, *Peep-yelp*, combine with 4 or more B-calls; only 1 B-call, i.e., *Peep-yelp*, combines with 4 or more A-calls. Applying this principle, a precise empirical condition quantifying linguistic productivity, we need, for rule-based productivity, at least 4 A-calls to combine with more than 3 B-calls, and conversely. Since we only find 2 A-calls combined with more than 3 B-calls, and only a single B-call combined with more than 3 A-calls, compositionality falls short of rule-based productivity. Overall, while interesting, the evidence for compositional meaning in animal communication systems seems extremely weak.<sup>2)</sup>

Girard-Buttoz et al. (2022) go even further and, assessing only “the potential for flexible meaning generation” rather than (compositional) meaning itself, argue that bigrams may recombine as units in more complex, hierarchically structured sequences of chimpanzee communication (Tai National Park, Ivory Coast). However, assuming full statistical independence as we did in the bonobo case, quantitative analysis demonstrates that these combinations also fail to result from a compositional process. Charles Yang [personal communication] found that without positional restrictions, the expected number of unique bigrams was never fewer than 80, i.e., a good deal more than the 58 unique bigrams produced from the 12 calls found in the study. No productivity for bigrams, no productivity for trigrams containing these bigrams as units. Once again, a finding about some *bounded* case of output does not tell us anything about the properties (ontogenetic or phylogenetic) of the human internal program which yields *unbounded* structured expressions.

Furthermore, acoustic identity is not sufficient for concluding rule-based composition. Bigram AB might contain elements that are acoustically identical – but communicatively unrelated – to A and B calls. The child-directed input may include *oar* (pronounced /ɔ:r/), *bit* (pronounced /bɪt/), and *orbit* (pronounced /'ɔ:r.bɪt/). The latter combines /ɔ:r/ and /bɪt/ without entailing compositional syntax. The word *orbit* is, in fact, a bisyllabic

lexical unit, not the result of a productive rule (as it is in *oarlock*, pronounced /'ɔ:r.la:k/). It has a different etymology but, obviously, the language learning child doesn't know and needn't know.

**4. Compositional meaning.** Human language is essentially an internal system for generating thought while animal communication is based on the principle that every internal representation is one-to-one associated with an organism-independent external event or internal state (Chomsky, 2013; Gallistel, 1990). The proposition *Man bites dog* contains the verbal predicate “bite” denoting a biting event involving two participants, a Patient argument, which it finds in the internal argument position, and an Agent argument, which it finds in the external argument position: [*Man* [*bites dog*]]. In *Combinatory Categorical Grammar*, such subcategorization is directly encoded in functor categories. The functional application  $(S \backslash NP) / NP \Rightarrow S \backslash NP$  follows a “cancellation rule” where a function, i.e., the category type  $(S \backslash NP) / NP$  for a transitive verb seeking an internal  $NP$  argument on its right, and the selected argument  $NP$  combine to yield the result  $S \backslash NP$ , canceling  $NP$  to leave  $S \backslash NP$ . The latter is the transitive verb phrase  $VP$  of phrase structure grammar. In turn,  $S \backslash NP$  takes external argument  $NP$  to yield a proposition  $S$ . The meaning of a complex expression is thus a compositional function of the meanings of its smaller parts and the grammatical rules that are used to combine them. Transitive verb “bite” takes an internal argument  $NP$  (Patient) “dog” to form a  $VP$  “bites dog,” which in turn takes an external argument  $NP$  (Agent) “man” to yield a propositional structure “man bites dog.” Merge Grammar imposes no linear order but theta theory associates the Patient role with internal argument (IA), and the Agent role with external argument (EA) in  $\{EA, \{V, IA\}\} = \{EA, VP\}$ , where  $V$  assigns Patient to IA and  $VP$  assigns Agent to EA. The man is doing the biting to the dog rather than conversely or something else. An unusual and unfamiliar meaning is legitimately and deterministically composed against all statistical odds. Compositional meaning clearly depends on structure, not on probability of event. The compositionality claim presupposes that we understand what the smaller parts mean and how the combinatorial rule works to derive the meaning of the whole from its constituent parts.

This notion of compositionality and the hierarchical syntactic structure presupposed are absent from animal communication systems. Little is known in the case of non-human primate “compositional meaning,” which today is hardly more than a flashy name for a black box solution. Primate calls (and call sequences) are somehow associated with external events in the primate natural environment that relate to danger, territoriality, mating, excitement, or food. That’s all we observe without understanding much. To understand compositional meaning in call sequences, assuming there is, we must understand the meanings of the unit calls and the rules used to (recursively) combine these to a more complex call sequence. However, composed meaning may not be compositional at all but instead context-sensitive, organism-specific, and based on pragmatic principles that apply to events in the animal’s natural environment (Schlenker et al., 2016). So far there’s only suggestive and misleading terminology without any of the explanatory insights we receive from investigating human language.

Today’s higher-order Markov chains suggested for animal vocalization (e.g., Bosshard et al., 2024; Le Floch et al., 2025) are equivalently accepted by string-based finite-state automata. These models finitely present the concatenation-based monoid algebraic structure of all animal call systems studied so far. However, the associativity property of the basic operation of finite-state models implies absence in animal communication of hierarchical structure necessary for non-trivial compositional meaning. Consequently, attempts to demonstrate hierarchical structure must be failures.<sup>3)</sup> It also follows that non-adjacent dependencies discussed in, e.g., Bosshard et al. (2024), Le Floch et al. (2025), Reber et al. (2019), or Watson et al. (2020) are misconceived. Summarizing, primate call sequences exhibit no hierarchical structure, no structure dependence, no recursion other than finite-state repeats (“loops”), and no non-trivial compositional meaning.

**5. Discussion.** The *Science* article (Berthet et al., 2025) adopts a distributional semantics view (“words with close meanings are used in similar contexts”) for compositional meaning. Word meanings captured in terms of ‘features of circumstances’ are converted into vectors and represented in a multi-dimensional semantic space in which meaning relations can be geometrically quantified. But statistical approximation is not theoretical explanation and has been shown to be irrelevant to predicting grammaticality. The class of finite-state Markov chains that produce higher-order statistical approximations to a language L do not come closer to matching the output of a (generative) grammar of L (Chomsky, 1956). And in distributional semantics we find a statistical model that at some descriptive level approximates meaning as the totality of contextual aspects that co-occur with the alert, much like the behaviorist definition of meaning as the sum of observable stimuli and responses. It does not provide a formal model that explains how compositional meaning is constructed from the meanings of bigram components. How could *pyow-hack*, the well-known ‘leopard-eagle’ alert of putty-nosed monkeys, be ‘compositionally’ scaled down to an ongoing or imminent group movement alert? It’s a reasonable question to ask (Schlenker et al., 2016).

For the meaning of AB to be compositionally derived from the meanings of A and B in distributional semantics, the Euclidian distance of AB from either A or B must be smaller than the distance between AB and any other call. As a consequence, in cases analogous Dutch *karwei* (“chore”), geometric distance metrics would successfully exclude compositional meaning for AB despite the acoustic identity of parts A and B with call units A and B. This is because the geometric distance between, e.g., *karwei* and *kar* (“cart”) or *wei* (“meadow”) is likely to be greater than that between *karwei* (“chore”) and *klus* (“job”). Therefore, these cases must, correctly it appears, be memorized rather than computed from the lexical items *kar* and *wei*.

Applying distributional methods to the bonobo vocal system, statistical analysis detected only three cases of non-trivial compositionality from a pool of 21 or – depending on positional restrictions – 42 potential bigrams. However, any such detection would be “accidental” since non-trivial compositionality requires hierarchical structure, which in turn requires the involvement of at least three elements. For purposes of efficient learning, memorization of these call sequences would be a far better strategy than rule learning (Yang, 2016), comparable to Dutch *karwei* (“chore”), memorized but not composed from lexical items *kar* and *wei*, which are otherwise involved in deriving compositional meaning from hierarchical structure, as in *{de-kar, {staat, {in, {de-wei}}}}* (“de cart is in the meadow”). More importantly, since bonobo compositionality is not hierarchical but string-based, there’s no comparison with human language. No theory is offered that *predicts* compositional meaning in some – but not in other – cases of primate vocalizations or *explains* how compositional meaning is formally derived from composed elements. Empirical evidence seems accidental rather than systematic. In particular, there’s no discussion of accidental gaps (unattested but possible call sequences) versus systematic gaps (unattested and impossible call sequences). The focus is on vocal behavior, probability distributions of attested calls, not on the internalized generative procedure underlying vocal behavior.

Furthermore, extrapolation to human language, where compositional meaning is recursive, yields incoherence. This is easily seen. Compositional meaning would incorrectly be ruled out, not just for cases like *let fall* since the semantic distance  $d(\text{let fall, drop})$  is likely to be smaller than  $d(\text{let fall, let})$ , but more generally also for analytical cases like *John convinced Mary that...*  $\Leftrightarrow$  *John caused Mary to believe that...* since  $d(AB, C) \ll d(AB, A)$  where  $A = \text{cause}$ ,  $B = \text{to believe that...}$ , and  $C = \text{convince that...}$ . Other cases come to mind. The geometric distance condition proposed would generally lead to system instability. This outcome is problematic for a position that suggests an evolutionary transition from non-human to human primate communication systems.

The structural ambiguities which are productive in human language and must exist by virtue of its hierarchical structure leave another critical question unanswered. The statistical method that is used computes meaning as a function of geometrical distance between word vectors. Can these kinds of methods be successfully employed to detect structural ambiguity in human language? Will they convincingly pass the litmus test of constrained ambiguity, hierarchical structure, and nontrivial composition? Not obvious at all. Experience, and therefore statistical information, is not relevant for the faculty of language, an internal program, although statistics can be an important factor in use (processing) and development (acquisition) as has been frequently argued, e.g., in Chomsky (2021). In fact, the distributional methods for segmentation and classification of structuralist linguistics (Harris, 1954) have been proven unable to offer explanation. This has become amply clear since the launch of Chomsky's generative research program.<sup>4)</sup> If distributional semantics works for primate vocalization but not for human language, how could 'compositional' meaning in animal communication be relevant for evolution of language? Apparently, the notion of (compositional) meaning has different uses in different domains, making comparison difficult if at all possible.

To conclude, this discussion illustrates "the unreasonable effectiveness of mathematics in the natural sciences" (Wigner, 1960), here biolinguistics, and demonstrates that the nontrivial composition results claimed for bonobos and other non-human primate species lead to immediate contradiction. The strings generated by Markov chains are finite-state computable by a concatenation operation whose associative property precludes generating the hierarchical structures implicit in non-intersective compositionality. Animal call systems lack the one operation needed to generate structured expressions, viz. recursive Merge (Berwick and Chomsky, 2016). Therefore, it is hard to see how (nontrivial) compositional meaning in human language could have evolutionary antecedents in animal call communication. We only seem to share with non-human primates a capacity to process sequential information. Primate calls may, at most, have intersective compositionality. But even here the conditional probabilities for productive compositionality cannot be established for primate bigrams by statistical relative frequencies. In sum, both qualitative considerations and quantitative tests argue against the existence of humanlike compositionality in the call systems of non-human primates.

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## Endnotes

1) Similarly, a *nuclear physicist* is “an expert in nuclear physics,” not “a physicist who is nuclear.” Interestingly, the intersective adjective in Dutch *nucleaire fysica* (“nuclear physics”) carries inflection (“-e”) that is suppressed from the non-intersective adjective in *nucleair fysicus* (“nuclear physicist”). This is the case generally. The property is explained if agreement signals interface interaction between syntax and morphology. Inflection is morpho-syntactic only. Since “nuclear physics” is internal to the lexical structure headed by the nominal suffix (“-ist”) in *[[nuclear physic-] -ist]*, the encapsulated adjective and noun are protected from syntactic agreement by derivational morphology. Non-intersective adjectives require hierarchical structure that prevents them from carrying agreement. We therefore get *nucleaire fysica* with inflection but *nucleair fysicus* without inflection since the derivational compound *[[nucleair fysic-] -us]* has the compound internal to the derivational morphology. Non-trivial compositionality requires hierarchical structure, which we argue is excluded from non-human primate communication.

2) If the bigrams are actually composed in a statistically independent way, the distribution of attested structures may be compared against the expected statistical distribution in yet another way. Since the tolerance threshold  $\theta_N$  here is  $42/\ln 42$  (Yang, 2016), rule  $R(A, B) = AB$  tolerates 11 “exceptions” (i.e., non-attested pairs) and must generate 31 unique bigrams (0.74) for productivity. Only 19 unique bigrams (0.45) were found. Far from a productive rule of composition. However, applicability of the *Tolerance Principle* to bonobo bigrams in this manner depends on whether the combinatorial rule, if even there is such a rule, applies freely to any pair of single calls (as in the analogous case of noun-noun compounds in human language). But, obviously, no one knows.

3) The inference from statistical probability of higher-order Markov Chains (or equivalent probabilistic finite-state automata) to hierarchical structure generated by rules of grammar is unwarranted, yielding incoherence.

4) As is well known since Chomsky discussed these cases in the sixties (Chomsky, 1964, 1965), the adjectives *eager*, *easy* and *certain* share the same context in *We love eager (easy, certain) victories* and are only deceptively similar in their distribution in cases like *John is easy (eager, certain) to please*. In fact, *certain* contrasts with *easy* and *eager* in *There's certain (\*eager, \*easy) to be a man in the house*, while *eager* differs from *certain* and *easy* in *John's eagerness (\*easiness, \*certainty) to please*, and *easy* behaves differently from both *eager* and *certain* in *John is easy (\*eager, \*certain) to work with* or *John is an easy (\*eager, \*certain) man to please*. Furthermore, only *easy* is excluded in *John is eager/certain/\*easy to come*, while only *certain* yields an ungrammatical result in *John is eager/easy/\*certain for me to please*, and only *eager* is ruled out in *It is easy/certain/\*eager to please*. Finally, there's the contrast between *\*John is easy for me to praise him* vs. *John is eager for me to praise him*. Conversely, there are cases of comparable meanings that must occur in different contexts. We get *John is likely* (vs. *\*probable*) *to come*, with comparable *It is likely (probable) that John comes*. Or *John is impossible* (vs. *\*possible*) *to work with*, with comparable *It's (im)possible to work with John*. The pair *He's pretty (funny) to look at* receives constrained alternatives in *\*It's pretty to look at him* vs. *It's funny to look at him*. Lastly, the predicates “eager” and “wise” in *She is eager (wise) to come* have different privileges of occurrence in *\*It's eager of her to come* vs. *It's wise of her to come*. Distributional semantics quantifies over geometrical distances between word vectors in semantic space. A vexing question is whether Euclidian distance alone can be successfully used to learn/predict the distributional differences between these superficially similar adjectives. It's not exactly a trivial task. This is something that must be shown to be the case, one way or another, not just assumed, before proceeding any further. [asterisk (\*) indicates ungrammaticality.]

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